4. Apply the sepia filter to the image.
5. Upload the modified image to S3.
6. Update the process state in the database.
7. Mark the message as done by deleting it from the queue.

Setting up the worker

To get started, you need some boilerplate code to load dependencies, initial AWS endpoints, and an endless loop to receive messages. The following listing explains the details.

Listing 16.5 Initializing the Imagery worker (worker/worker.js)

```
const AWS = require('aws-sdk');                                    ①
const assert = require('assert-plus');
const Jimp = require('jimp');
const fs = require('fs/promises');


const db = new AWS.DynamoDB({});                                   ②
const s3 = new AWS.S3({});
const sqs = new AWS.SQS({});

const states = {
  'processed': processed
};

async function processMessages() {                                 ③
  let data = await sqs.receiveMessage({                            ④
    QueueUrl: process.env.ImageQueue,
    MaxNumberOfMessages: 1
  }).promise();
  if (data.Messages && data.Messages.length > 0) {
    var task = JSON.parse(data.Messages[0].Body);
    var receiptHandle = data.Messages[0].ReceiptHandle;
    assert.string(task.imageId, 'imageId');                        ⑤
    assert.string(task.desiredState, 'desiredState');
    let image = await getImage(task.imageId);                      ⑥
    if (typeof states[task.desiredState] === 'function') {
      await states[task.desiredState](image);                      ⑦
      await sqs.deleteMessage({                                    ⑧
```